

The Fertility, Marriage, and Gender Equality Quandary

Anson Zhou

The University of Hong Kong

September 2026

Motivation

- Three demographic transformations over the past half-century:
 1. Falling fertility, often well below replacement
 2. Declining dual parenthood: rising single-parent households
 3. Converging gender income gaps (Goldin 2014)

Motivation

- Three demographic transformations over the past half-century:
 1. **Falling fertility**, often well below replacement
 2. **Declining dual parenthood**: rising single-parent households
 3. **Converging gender income gaps** (Goldin 2014)
- Each motivates a distinct policy agenda
 1. Pro-fertility: pension sustainability, growth (Jones 2022)
 2. Pro-marriage: children's outcomes (Autor et al. 2019)
 3. Gender equality: reducing misallocation (Hsieh et al. 2019)

Motivation

- Three demographic transformations over the past half-century:
 1. **Falling fertility**, often well below replacement
 2. **Declining dual parenthood**: rising single-parent households
 3. **Converging gender income gaps** (Goldin 2014)
- Each motivates a distinct policy agenda
 1. Pro-fertility: pension sustainability, growth (Jones 2022)
 2. Pro-marriage: children's outcomes (Autor et al. 2019)
 3. Gender equality: reducing misallocation (Hsieh et al. 2019)
- These agendas are discussed separately—yet the underlying decisions are closely connected

This Paper

1. Document a **three-way trade-off** in 115 countries; same magnitude in OECD & non-OECD but **different dominant tensions**
2. Develop an **equilibrium marriage-market model**: marriage bundles income transfer with joint fertility
3. Show level policies improve two at the expense of the third; **childcare redistribution** can improve all three under bilateral conditions
4. Calibrate to Mexico 1990–2025; intergenerational extension

Related Literature

- Fertility & female labor supply through time cost of children (Erosa et al. 2016, Blundell et al. 2016; Feyrer et al. 2008)
→ Place inside marriage market where the marital contract adjusts
- Marriage-market models (Choo & Siow 2006, Chiappori et al. 2017)
→ Marriage bundles income sharing with joint fertility
- Gender revolution: men's domestic roles restabilize families (Goldscheider et al. 2015, Doepke & Kindermann 2019)
→ Identify bilateral conditions under which this conjecture holds
- Technology & the family (Greenwood et al. 2005, Ngai & Petrongolo 2017)
→ Decompose into gender-neutral and gender-biased components

Motivating Facts

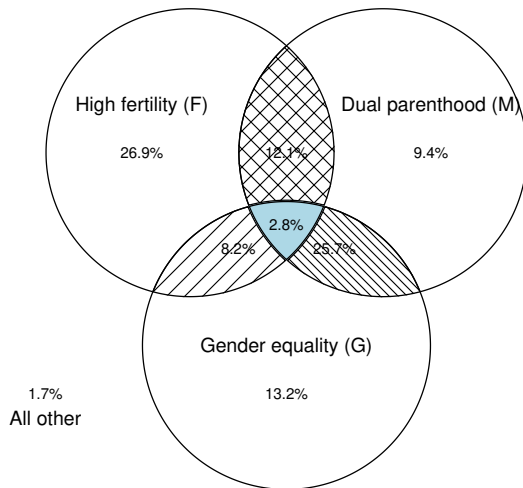
Data Construction

Sample	Countries	Obs	Dual parenthood source
Full	115	1,764	IPUMSI + OECD Family Database
OECD	39	879	OECD Family Database (+ IPUMSI)
Non-OECD	76	885	IPUMSI only

- Fertility: total fertility rate (WDI)
- Dual parenthood: share of children living with both parents
- Gender equality: female share of labor income (WID)
- Construct dummies at sample-specific medians: high F, dual M, equal G
- **Dartboard benchmark**: under independence, trinity share $\approx 0.5^3 = 12.5\%$

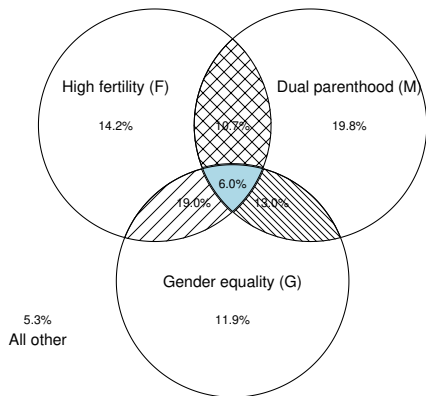
The Trinity is Rare: Full Sample

bootstrap

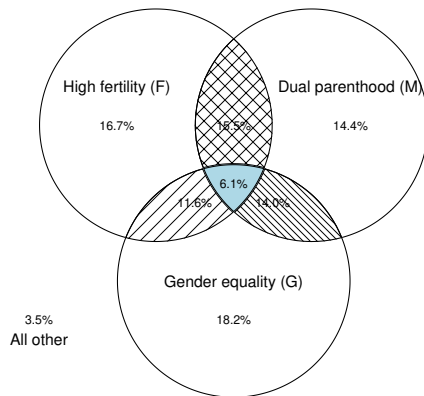


Same Shortfall, Different Sources

bootstrap



OECD ($N = 879$, 39 countries)



Non-OECD ($N = 885$, 76 countries)

Remarkably similar magnitude—but the Venn patterns differ

Decomposing the Shortfall

bootstrap

$$\underbrace{p_{111} - p_{FpMpG}}_{\text{shortfall}} = \underbrace{\delta_{FM} \cdot p_G}_{F-M} + \underbrace{\delta_{FG} \cdot p_M}_{F-G} + \underbrace{\delta_{MG} \cdot p_F}_{M-G} + \varepsilon$$

	OECD	Non-OECD
F-M	63.4%	26.7%
F-G	-1.1%	55.9%
M-G	45.1%	37.9%
Residual	-7.4%	-20.5%

- $\delta_{XY} < 0$: X, Y repel each other
- OECD: F-M tension dominates; F-G $\approx$ zero
- Non-OECD: F-G tension dominates

Questions for Theory

1. Why do the **dominant tensions differ** across OECD and non-OECD countries?
2. Can policy **loosen** the three-way trade-off?
3. What drives the demographic transition **quantitatively**?

Theory

Basic Setup

- Building on Becker (1973), Choo and Siow (2006)
- Individuals with gender $g \in \{\sigma, \varphi\}$ and preference

$$u(c, n) = \left((1 - \beta) \cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (1)$$

where $\rho > 1$ so that $u(c, 0)$ is well-defined

- Homogeneous wage within gender: w^{σ} and w^{φ}
- Gender wage gap: $\Gamma^w = w^{\sigma} / w^{\varphi}$

Marriage and Fertility – Men

- If single, men consume labor income but have no children

$$V^{\sigma^{\rightarrow},s} = u(w^{\sigma^{\rightarrow}}, 0) \quad (2)$$

- Once married, husbands transfer α share of income to wives

$$V^{\sigma^{\rightarrow},m} = \lambda \cdot u((1 - \alpha)w^{\sigma^{\rightarrow}}, n^m) \quad (3)$$

→ $\alpha \in (-w^{\sigma^{\rightarrow}}/w^{\sigma^{\rightarrow}}, 1)$ is endogenous in equilibrium

→ $\lambda > 0$ is the common benefit of marriage (norms, companionship)

Marriage and Fertility – Women

- Single women solve

$$V^{\text{♀},s} = \max_{c,l,n} u(c,n) \quad \text{s.t.} \quad c = w^{\text{♀}}l, \quad l = 1 - \chi n \quad (4)$$

- Married women solve

$$V^{\text{♀},m} = \max_{c,l,n} \lambda \cdot u(c,n) \quad \text{s.t.} \quad c = \alpha w^{\text{♂}} + w^{\text{♀}}l, \quad l = 1 - \chi n \quad (5)$$

- Trade-off between consumption and fertility through female labor supply
- In the baseline, wives bear all childcare ($\chi^{\text{♂}} = 0$)

Marriage Market Equilibrium

- Idiosyncratic taste shock ϵ for marriage, Fréchet with shape θ
- Marriage rates:

$$\mathcal{M}^g = \frac{(V^{g,m})^\theta}{(V^{g,m})^\theta + (V^{g,s})^\theta} \quad (6)$$

- Equilibrium: $\mathcal{M}^{\sigma^\heartsuit} = \mathcal{M}^{\sigma^\spadesuit} = \mathcal{M} \iff \frac{V^{\sigma^\heartsuit,m}}{V^{\sigma^\heartsuit,s}} = \frac{V^{\sigma^\spadesuit,m}}{V^{\sigma^\spadesuit,s}}$
- Two-dimensional price (α, n^m) : marriage bundles income transfer *and* marital fertility

Marriage Market Equilibrium

- Idiosyncratic taste shock ϵ for marriage, Fréchet with shape θ
- Marriage rates:

$$\mathcal{M}^g = \frac{(V^{g,m})^\theta}{(V^{g,m})^\theta + (V^{g,s})^\theta} \quad (6)$$

- Equilibrium: $\mathcal{M}^{\sigma^\uparrow} = \mathcal{M}^{\varnothing} = \mathcal{M} \iff \frac{V^{\sigma^\uparrow,m}}{V^{\sigma^\uparrow,s}} = \frac{V^{\varnothing,m}}{V^{\varnothing,s}}$
- **Two-dimensional price** (α, n^m) : marriage bundles income transfer *and* marital fertility
- Lemma 1: Unique interior equilibrium; α^* and n^m independent of λ, θ
- Lemma 2: Gains from marriage: $V^{\varnothing,m}/V^{\varnothing,s} = \lambda \cdot (1 + \alpha^* \Gamma^w)$,
 $n^m/n^s = 1 + \alpha^* \Gamma^w > 1$

Three-Way Trade-Off: Intuition

- Equilibrium conditions reveal fundamental tension:

$$\mathcal{M} = \frac{[\lambda \cdot (1 + \alpha^* \Gamma^w)]^\theta}{1 + [\lambda \cdot (1 + \alpha^* \Gamma^w)]^\theta} \quad (7)$$

$$\Gamma^y = \Gamma^w / l^\varnothing \quad (8)$$

$$l^\varnothing = 1 - \chi n \quad (9)$$

- Insight:** Higher n reduces $l^\varnothing$, requiring low Γ^w for gender equality—but low Γ^w reduces the income-sharing attraction of marriage ($\alpha^* \Gamma^w$), lowering $\mathcal{M}$

Why Tensions Differ: The Wage Gap

Lemma 3: Γ^w determines which pairwise tension dominates

High Γ^w (Non-OECD)

- (i) F–G amplification *strong*:
 $\Gamma^y = \frac{\Gamma^w}{1-\chi n}$, even moderate n
pushes Γ^y above equality
- (ii) F–M coupling *tight*: large marital
fertility premium $q \gg 1$ ties
marriage and fertility together

⇒ F–G tension dominates

Low Γ^w (OECD)

- (i) F–G amplification *weak*:
moderate n compatible with
equality
- (ii) F–M coupling *loose*: $q \approx 1$, so
marriage and fertility decouple
across countries

⇒ F–M tension dominates

Matches decomposition: 55.9% F–G in non-OECD; 63.4% F–M in OECD

Policy Effects

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{\text{♀}}$)	−	−	− (narrows)
Reduce child costs ($\downarrow \chi$)	+	+	+ (widens)

- No single *level* policy can improve all three outcomes
- Trade-off binds because
 1. Wives' childcare burden ties fertility to gender gaps through $l^{\text{♀}}$
 2. Income-sharing motive and composition ($n^m > n^s$) tie fertility to marriage

Policy Effects

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{\text{♀}}$)	−	−	− (narrows)
Reduce child costs ($\downarrow \chi$)	+	+	+ (widens)

- No single *level* policy can improve all three outcomes
- Trade-off binds because
 1. Wives' childcare burden ties fertility to gender gaps through $l^{\text{♀}}$
 2. Income-sharing motive and composition ($n^m > n^s$) tie fertility to marriage
- Proposition: Combining $\uparrow w^{\text{♀}}$ with $\uparrow \lambda$ can improve all three (continuity argument), but requires **delicate calibration**—motivates a simpler, single-instrument approach

Redistributing Childcare

Model with Husband-Provided Childcare

- Extension: $\chi^{\text{♀}}$ for wives, $\chi^{\text{♂}}$ for husbands, with $\chi^{\text{♀}} + \chi^{\text{♂}} = \chi$
- **Paternal care efficiency** $\eta \in (0, 1]$: father's raw time per child = $\chi^{\text{♂}} / \eta$
→ $\eta = 1$: equal efficiency; $\eta < 1$: father needs more time per unit of care
- Wife's effective childcare cost:

$$\tilde{\chi}^{\text{♀}} = \frac{w^{\text{♀}}\chi^{\text{♀}} + \alpha w^{\text{♂}}\chi^{\text{♂}}/\eta}{w^{\text{♀}} + \alpha w^{\text{♂}}} \quad (10)$$

- Income-weighted average: a child uses some of her earning capacity and some of his (deflated by η)
- Consider shifting δ effective units of childcare from wife to husband

Conditions for Joint Improvement

Proposition: Redistribution raises n , $\mathcal{M}$, and narrows Γ^y if:

1. Wife's effective cost falls: $w^{\text{♀}} > \alpha w^{\text{♂}} / \eta$

Her freed earnings exceed the transfer-base loss from his care

2. Income convergence after fertility adjusts:

Childcare redistribution more than compensates for additional care from higher n

3. Husband gains from adjusted contract:

Despite more caregiving, the additional fertility is valuable enough

Conditions for Joint Improvement

Proposition: Redistribution raises n , $\mathcal{M}$, and narrows Γ^y if:

1. Wife's effective cost falls: $w^{\varnothing} > \alpha w^{\sigma} / \eta$

Her freed earnings exceed the transfer-base loss from his care

2. Income convergence after fertility adjusts:

Childcare redistribution more than compensates for additional care from higher n

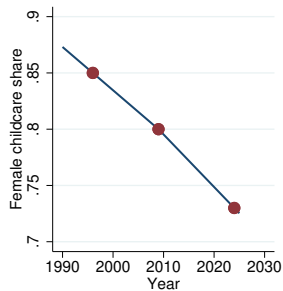
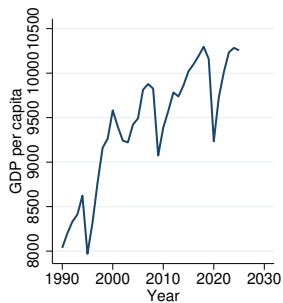
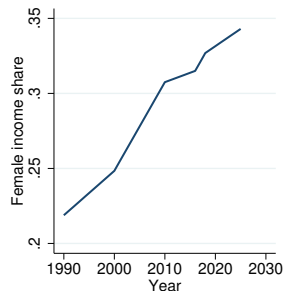
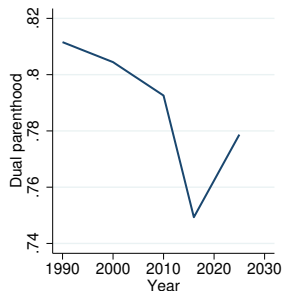
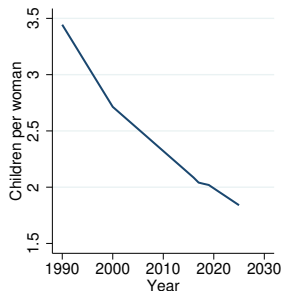
3. Husband gains from adjusted contract:

Despite more caregiving, the additional fertility is valuable enough

- All three conditions tighten as η falls $\implies$ **minimum threshold $\bar{\eta}$**
- Connects to Goldscheider et al. (2015): restabilization requires *both* spouses to gain

Quantitative Analysis

Mexico's Transition (1990–2025)

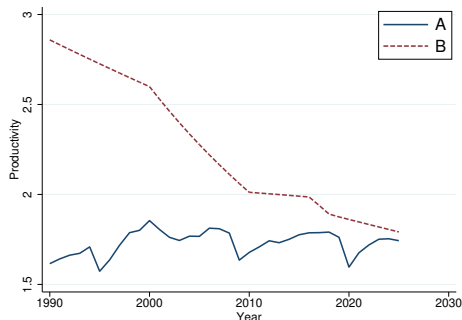


Calibration

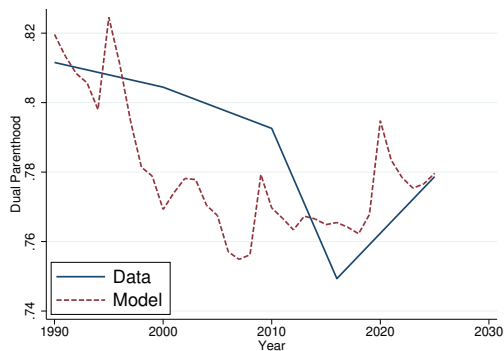
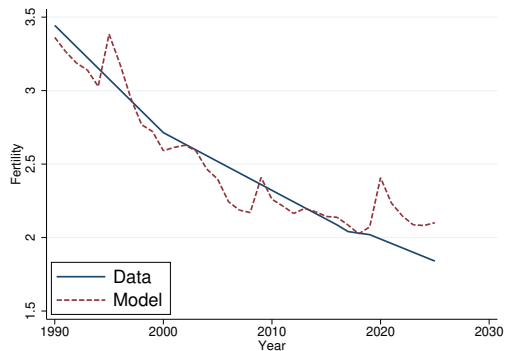
[Shapley shares](#)[CE table](#)

- Three time-varying inputs:
 - $A_t = w_t^{\sigma}$: gender-neutral productivity
 - $B_t = w_t^{\sigma} / w_t^{\varphi}$: relative wage gap
 - s_t : female childcare share (ENUT 1996, 2009, 2024)
- Fix $\chi = 0.075$, $\eta = 1$. Calibrate 4 parameters to fertility & dual parenthood:

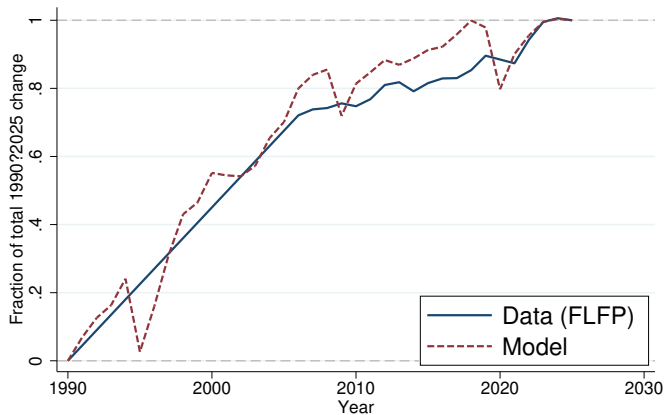
	Value	Target
β	0.077	Fertility level
ρ	2.50	Fertility trend
λ	0.954	Marriage level
θ	12.0	Marriage trend



Model Fit: Fertility and Dual-Parenthood

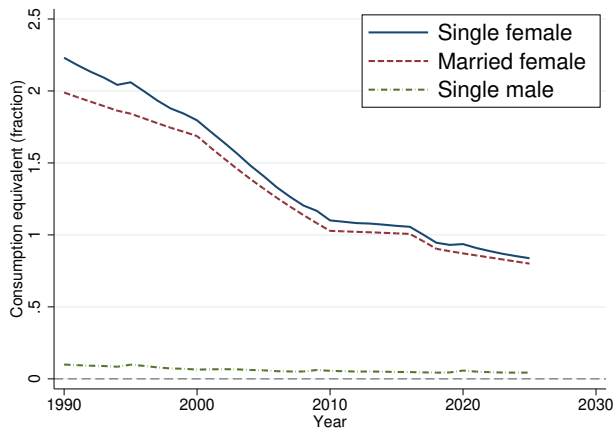


Non-Targeted: Female Labor Supply



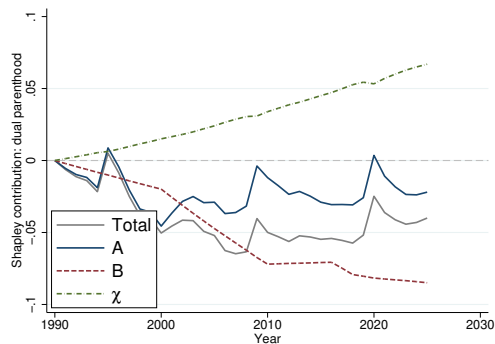
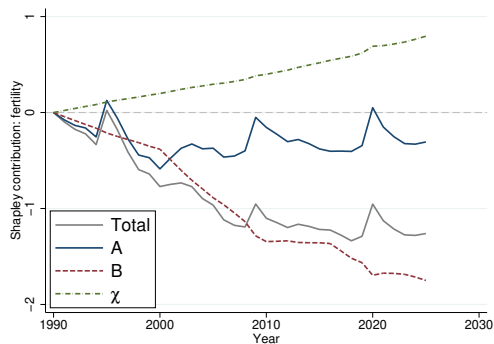
Both series: 0 in 1990, 1 in 2025
Correlation = 0.979
FLFP is not used in calibration

Relative Welfare



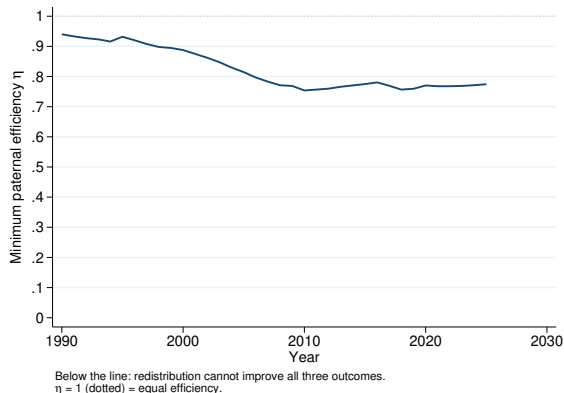
- All agent types converge toward married-male welfare

Shapley Decomposition

[full table](#)

- Declining gender wage gap B_t is the **dominant driver**: 139% of fertility decline, 95% of income convergence, 97–107% of welfare convergence
- More equal childcare χ **partially buffers** fertility and dual parenthood

Scope for Redistribution Widens Over Time

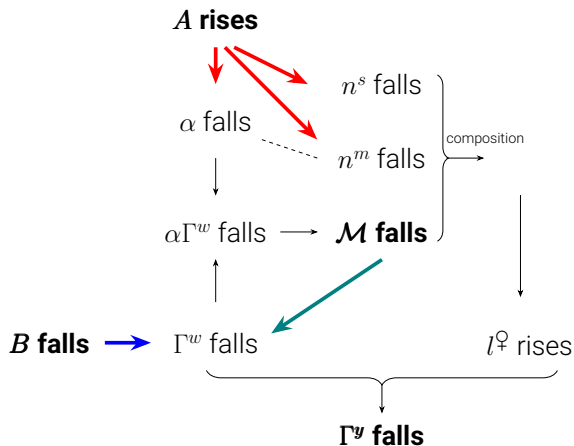


- $\bar{\eta}$ declines from 0.94 (1990) to 0.77 (mid-2010s): fathers could be 23% less efficient and redistribution still improves all three outcomes
- Binding constraint throughout: Condition 3 (husband's participation)

Dynamic Extension

Dynamic Propagation

- Boys more sensitive to father absence (Autor et al. 2019, Wasserman 2020). Law of motion: $\Gamma_{t+1}^w = \mathcal{H}(\mathcal{M}_t)$, $\mathcal{H}' > 0$



Red: A_t rises Blue: B_t falls Teal: $\mathcal{M}_t \rightarrow \Gamma_{t+1}^w$ feedback

Conclusion

- Three-way trade-off in 115 countries; Γ^w determines which tension binds—F–M in OECD, F–G in non-OECD
- Equilibrium marriage-market model: marriage bundles income transfer with fertility, generating the trade-off endogenously
- Level policies trade one outcome against others; childcare redistribution can improve all three when both spouses gain ($\bar{\eta}$: 0.94 $\rightarrow$ 0.77)
- Mexico: declining wage gap drives the transition; more equal childcare partially buffers fertility and marriage
- Broader lesson: both sides of the marriage market must gain for family policy to succeed

Bootstrap Inference

[back](#)

	Full		OECD		Non-OECD	
	Est.	SE	Est.	SE	Est.	SE
Trinity share	0.028	0.014	0.060	0.020	0.061	0.023
Shortfall (pp)	−9.65	1.40	−6.34	2.04	−6.38	2.30
$I(F; M)$ (bits)	0.121	0.054	0.076	0.054	0.014	0.025
$I(F; G)$ (bits)	0.239	0.066	0.000	0.017	0.060	0.044
$I(M; G)$ (bits)	0.015	0.018	0.038	0.037	0.027	0.033

1,000 replications clustered by country.

Decomposition Bootstrap

[back](#)

Component	OECD		Non-OECD	
	Point (pp)	p -value	Point (pp)	p -value
F–M tension	–4.02	0.003	–1.71	0.253
F–G tension	+0.07	0.964	–3.57	0.005
M–G tension	–2.86	0.044	–2.42	0.105
Three-way	+0.47	0.706	+1.31	0.300
Total	–6.34		–6.38	

- F–M tension: significantly negative in OECD ($p = 0.003$)
- F–G tension: significantly negative in non-OECD ($p = 0.005$), ≈ 0 in OECD ($p = 0.964$)

Shapley Shares at 2025

[back](#)

Outcome	A (%)	B (%)	χ (%)
Fertility	24.3	138.8	-63.1
Dual parenthood	55.0	212.5	-167.4
Gender income gap	3.3	94.5	2.2
CE: Single female	2.4	97.7	-0.2
CE: Married female	0.5	107.2	-7.6
CE: Single male	21.8	82.6	-4.4

% of total change from 1990 baseline. Negative share = force moves outcome in opposite direction.

Consumption Equivalents at Selected Years

[back](#)

Year	α	CE: Single ♀	CE: Married ♀	CE: Single ♂
1990	0.044	2.23	1.99	0.10
1995	0.044	2.06	1.84	0.10
2000	0.034	1.80	1.69	0.06
2005	0.036	1.41	1.32	0.06
2010	0.038	1.10	1.03	0.06
2015	0.033	1.06	1.01	0.05

Benchmarked to married male (CE = 0). Positive values = agent is worse off.
Married-female CE halves from 1.99 to 1.01; driven by rising $w^{\text{♀}}$, not larger transfers.